

Hill Hold Control System for Electric Vehicles Using ESP32 and IMU Sensor

Problem Statement

When an electric vehicle stops on an uphill road due to traffic, the rider usually applies the brake to prevent the vehicle from rolling backward.

However, when the rider releases the brake and attempts to accelerate:

- The vehicle may roll backward.
- The rider may lose balance.
- The vehicle may collide with vehicles behind.
- It creates safety concerns, especially on steep slopes.

Therefore, a system is required to automatically hold the vehicle in position on an inclined road until the rider starts moving forward.

Project Objective

The objectives of this project are:

- Detect when the vehicle is on an uphill slope.
- Detect when the vehicle is stationary.
- Prevent the vehicle from rolling backward.
- Automatically release the hold when acceleration is applied.
- Improve rider safety and comfort.

Working Principle

The system continuously monitors:

- Vehicle inclination angle
- Vehicle speed
- Throttle position
- Brake status

When:

1. The vehicle is on an uphill slope.
2. Speed becomes zero.
3. Brake is applied.

The system enters **Hill Hold Mode**.

Even if the rider releases the brake, the system keeps the motor energized with a small holding torque to prevent rollback.

When the rider rotates the throttle:

- Hill Hold is released.
- The vehicle moves forward smoothly.

Components Required

Component	Quantity
ESP32	1
MPU6050 (Accelerometer + Gyroscope)	1
Hall Effect Speed Sensor	1
Motor Driver / ESC	1
BLDC Motor	1
OLED Display	1
Brake Switch	1
Throttle Sensor	1
Battery Pack	1

Why MPU6050 is Used

The MPU6050 measures:

- Vehicle tilt angle
- Acceleration
- Orientation

Using accelerometer data, the ESP32 can determine whether the vehicle is on:

Road Type	Angle
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Flat Road	0°–3°
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Mild Slope	3°–7°
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Steep Slope	>7°
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If the angle exceeds a predefined value, the system identifies an uphill condition.

Detection Logic

Assume:

Slope Angle > 7°

Vehicle Speed = 0 km/h

Brake Applied = TRUE

Then:

Hill Hold Mode = ON

System Operation

Scenario

A rider reaches traffic on an uphill road.

Vehicle speed:

20 km/h → 10 km/h → 0 km/h

The brake is applied.

The MPU6050 detects:

Slope Angle = 12°

The ESP32 activates Hill Hold Mode.

Now:

- Rider releases the brake.
- Vehicle remains stationary.
- No rollback occurs.

When the rider accelerates:

- Throttle input detected.
- Hold released.
- Vehicle moves forward.

Control Strategy

The ESP32 continuously checks:

IF

Slope > 7°

AND

Speed = 0

AND

Brake Applied

THEN

Activate Hill Hold

Hold Release Logic

IF

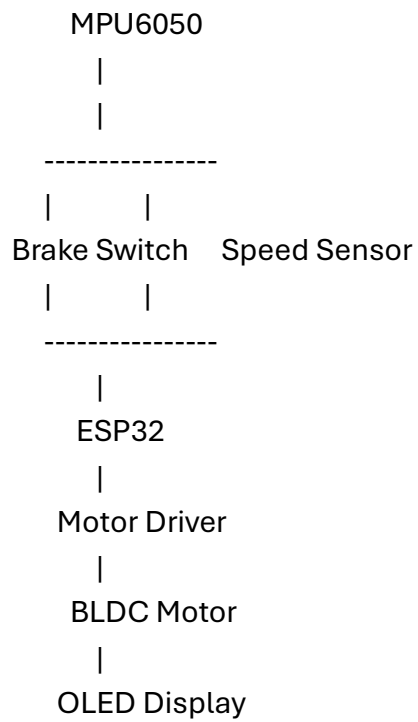
Throttle > 10%

THEN

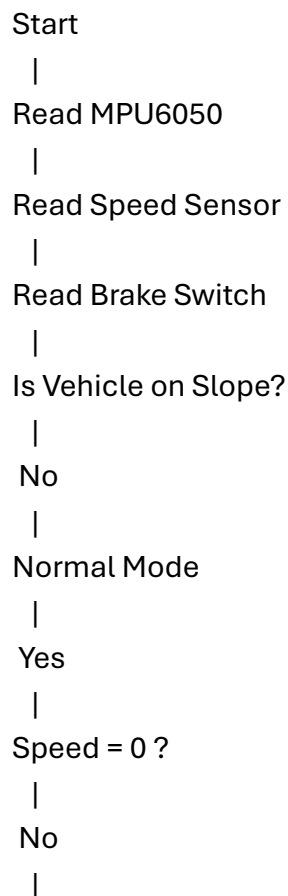
Release Hill Hold

Allow Motor Rotation

System Architecture



Flowchart



Normal Mode

|

Yes

|

Brake Applied?

|

No

|

Normal Mode

|

Yes

|

Activate Hill Hold

|

Monitor Throttle

|

Throttle Applied?

|

No

|

Keep Holding

|

Yes

|

Release Hold

|

Move Forward

OLED Display Output

Normal Mode

Speed : 25 km/h

Slope : 2°

Status : Normal

Hill Hold Active

Speed : 0 km/h

Slope : 12°

HILL HOLD

ACTIVE

Vehicle Moving

Throttle Applied

Hill Hold Released

Moving Forward

Implementation Methods

Method 1 (Recommended for College Project)

Electronic Motor Holding

Instead of using a mechanical brake:

- ESP32 commands the motor controller.
- A small holding torque is generated.
- Motor resists backward movement.

Advantages:

- Easy implementation.
- Low cost.
- Similar to actual EVs.

Method 2 (Industrial Method)

Electromagnetic Brake

When Hill Hold activates:

- Electromagnetic brake locks the wheel.
- Brake releases automatically when throttle is applied.

Advantages:

- Very strong holding force.

- Used in industrial vehicles and elevators.

Disadvantage:

- Higher cost.

Additional Features

Rollback Warning

If rollback begins:

WARNING

ROLL BACK

DETECTED

Buzzer activates.

Hill Hold Timer

Display:

Hill Hold Time

5 Seconds

Conclusion

The **Hill Hold Control System using ESP32, MPU6050, Speed Sensor, and Motor Controller** prevents an electric vehicle from rolling backward on inclined roads. By detecting slope angle and vehicle stoppage conditions, the system automatically generates holding torque and keeps the vehicle stationary until the rider accelerates. This feature improves safety, comfort, and driving confidence, making it an excellent embedded systems and automotive automation project for an Automation and Robotics Engineering student.